No tenth discriminant

A modern reconstruction using tools documented by 1830

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Abstract

We prove that a negative fundamental discriminant of class number one has absolute value $q < 32768$, and then exclude every possibility outside $3, 4, 7, 8, 11, 19, 43, 67, 163$ by explicit residue tables. The proof uses binary quadratic forms, elementary calculus and prime estimates, determinant expansion, and arithmetic modulo an irreducible quadratic congruence. These foundations were available in print by 1830. Ideal theory and Hadamard's determinant inequality are replaced by elementary arguments. The tables and all numerical inequalities can be checked from the paper; software is supplementary. We distinguish the historical sources documenting those tools from the 2026 sources to which we owe the argument. The proposed 1830 cutoff uses the interpretation of "results and concepts" stated in Section 1.1. We do not claim that the proof was known then or that 1830 is the earliest possible year.

1 The historical question and the theorem

Tao, reporting a question of Ben Green [M6], asks how early one could have proved the nonexistence of a tenth discriminant using only results and concepts already in the literature. We adapt the elementary companion [M7] to this question. The essential changes are to replace ideal theory by explicit quadratic forms, replace Hadamard's inequality by the permutation expansion of a determinant, and replace a computer sieve by a printed certificate. The proof below is complete independently of that companion. Contemporary sources are cited for attribution and context, with the convention explained below.

Throughout, a *negative fundamental discriminant* means an integer $D < 0$ satisfying either: D is squarefree and $D \equiv 1 \pmod 4$; or $D = 4d$, where d is squarefree and $d \equiv 2$ or $3 \pmod 4$. Put $q = -D$. We use the original quadratic-form formulation of the class-number problem: $h(D)$ is the number of proper equivalence classes of primitive positive definite forms

$$Ax^2 + Bxy + Cy^2, \qquad B^2 - 4AC = D.$$

Here primitive means $\gcd(A, B, C) = 1$, and proper equivalence means an integral change of variables of determinant one. Equivalent forms represent exactly the same integers. In particular, if $h(D) = 1$, every such form is equivalent to the principal form. We will need no composition law for forms.

Theorem 1. *If $D < 0$ is fundamental and $h(D) = 1$, then $q < 32768 = 2^{15}$.*

Theorem 2. *For a negative fundamental discriminant D,*

$$h(D) = 1 \quad \Longleftrightarrow \quad |D| \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

1.1 Historical scope and citation conventions

The mathematical statements above and the proposed historical cutoff are separate claims. Our interpretation of the challenge permits new constructions and deductions using objects, operations and results documented by 1830. It does not permit importing a later concept merely because its definition or a theorem about it can be written in elementary terms.

In particular, we treat the finite array of polynomial derivatives in Section 5 as a new construction from determinants, polynomials, derivatives and quadratic congruences. We exhibit that array and prove its nonvanishing, size and divisibility properties directly. We assume no general theorem about interpolation determinants. The strategic use of this construction is modern; our historical claim concerns the concepts needed to formulate and justify it. Whether that distinction meets the challenge's intended meaning of "concepts" is a separate historical judgement.

References labelled H document the availability of the foundations. References labelled M identify modern sources of the argument or the context of the question. Attributions beside the proof appear in footnotes headed *Modern attribution*; they give credit without serving as substitutes for proofs. Every mathematical assertion taken from a modern source and needed here is proved in the text. Removing the modern attribution notes would therefore leave the mathematical argument complete, although that test alone would not establish its historical eligibility. We make no claim that the construction was known in 1830, or that it could not be reconstructed from earlier tools.

1.2 Historical foundations

Gauss's 1801 treatment [H1, Section V] supplies the language of forms, integral substitutions, representation and reduction. We use the modern middle-coefficient convention B, whereas Gauss writes $2b$. For odd B, double our whole form to obtain his integral triple $(2A, B, 2C)$; for even B, use $(A, B/2, C)$. Scaling all forms in a fixed case preserves proper equivalence. The squarefreeness condition in our definition is also an explicit arithmetic restriction, not an input from ideal theory. Cauchy's 1821 text [H2, Chapters III, VI–VIII] documents determinant algebra, series and complex absolute values; ordinary differential and integral calculus is also in his 1823 lectures [H3]. Galois's published 1830 paper [H4] explicitly treats arithmetic with roots of irreducible polynomial congruences, p-power conjugation, and differentiation of those polynomials. Only the quadratic case is used here, and it is proved below.

The resulting dependency dates are summarized in Table 1. The terms "field" and "Frobenius" abbreviate the explicit quadratic arithmetic explained in Section 2; no later structural theorem is required. All estimates for primes are derived from factorials and binomial coefficients. In particular, we do not use an externally supplied prime-distribution theorem.

Foundation used	Pre-1831 source or explicit reduction
Forms, congruences, integer factorization	Gauss (1801), Sections I–V
Determinants and linear equations	Cauchy (1821), Chapter III
Complex moduli, series, elementary calculus	Cauchy (1821), Chapters VI–VIII; lectures (1823)
Arithmetic modulo an irreducible quadratic; polynomial derivatives	Galois (1830); elementary details in Section 2
Prime estimates, determinant estimates and finite exclusion	Proved here from those foundations

Table 1: Dates concern available foundations, not discovery of the present proof.

2 Arithmetic with forms and quadratic congruences

2.1 The small-prime obstruction without ideals

We first establish the small-prime obstruction directly from quadratic forms.[1] First suppose $D = -4m$ is even and fundamental, so m is squarefree and $m \equiv 1$ or $2 \pmod 4$. The principal form is $x^2 + my^2$. If $m \equiv 2 \pmod 4$, the form with coefficients $(2, 0, m/2)$ is primitive and has discriminant $-4m$. If $m \equiv 1 \pmod 4$, use $(2, 2, (m+1)/2)$ instead. Both represent 2. If $h(D) = 1$, so must $x^2 + my^2$, which is impossible when $m > 2$. Thus $q > 8$ forces q to be odd.

Now q is squarefree and $q \equiv 3 \pmod 4$. Put

$$u = \sqrt{q}, \quad \omega = (1 + iu)/2, \quad Q(a, b) = |a + b\omega|^2 = a^2 + ab + \frac{q+1}{4}b^2 = (a + b/2)^2 + qb^2/4. \tag{1}$$

This is the principal form of discriminant $-q$. If p is an odd prime and $-q$ is a square modulo p (including zero), choose an odd integer B with $B^2 \equiv -q \pmod p$. Then

$$\left(p, B, \frac{B^2 + q}{4p}\right)$$

is an integral positive definite form of discriminant $-q$, representing p. It is primitive: when $p \nmid q$, we have $p \nmid B$; when $p \mid q$, squarefreeness gives $p \nmid (B^2 + q)/(4p)$. Consequently $h(D) = 1$ forces $p = Q(a, b)$. Here $b \neq 0$, since a prime is not a square, and hence $p \geq q/4$.

If $q \equiv 7 \pmod 8$, the primitive form $(2, 1, (q+1)/8)$ represents 2. The same reasoning contradicts $q > 8$. We have therefore proved, for every $q > 8$ of class number one,

$$q \equiv 3 \pmod 8, \qquad p < q/4, p \text{ odd prime} \implies -q \text{ is a nonzero nonsquare modulo } p. \tag{2}$$

This statement alone will suffice for the finite certificate.

For Theorem 1, assume from now until Section 7 that $h(D) = 1$ and $q \geq q_0 := 32768$. Then q is prime: a composite squarefree q would have a prime divisor $p \leq \sqrt{q} < q/4$, contrary to (2). For this prime q, we call an odd prime $p \neq q$ *split* or *inert* according as $-q$ is a square or nonsquare modulo p. We also call 2 inert, and call q ramified. These are just names for the three congruence cases. All noninert primes are at least $q/4$.

2.2 The required arithmetic modulo an inert prime

Let $\mathcal{A} = \mathbb{Z} + \mathbb{Z}\omega$. The identities

$$\omega^2 = \omega - (q+1)/4, \qquad \bar{\omega} = 1 - \omega$$

show directly that $\mathcal{A}$ is closed under addition, multiplication and conjugation. Every nonzero $z = a + b\omega$ has positive integer norm $z\bar{z} = Q(a, b)$, hence $|z| \geq 1$. We write $z \equiv z' \pmod p$ when both coefficients of $z - z'$ are multiples of p; this definition respects addition and multiplication.

For inert p, the polynomial $f(X) = X^2 - X + (q+1)/4$ is irreducible modulo p: for odd p its discriminant is the nonsquare $-q$; for $p = 2$ it is $X^2 + X + 1$. Arithmetic with the p^2 expressions $a + b\omega$, a, b taken modulo p, is thus a field. Indeed the polynomial Euclidean algorithm gives an inverse modulo f for every nonzero polynomial of degree at most one. This is exactly the kind of arithmetic described by Galois in 1830 [H4]; it does not require ideals or abstract field theory.

[1] *Modern attribution.* The obstruction was emphasized by Silberman [M5]; the derivation here uses explicit forms in place of ideal theory.

In this arithmetic,

$$z^p = \bar{z} \quad \text{for every } z \in \mathcal{A}. \tag{3}$$

To prove this, the binomial theorem modulo p and Fermat's congruence show that ω^p is a root of f. It cannot equal ω: the polynomial $X^p - X$ already has all p scalar residues as roots, and a nonzero polynomial of degree p over a field has at most p roots. (Successive division by $X - a$ proves this root count.) Therefore $\omega^p = 1 - \omega$, the other root of f. Applying the binomial theorem once more proves (3). This proof includes $p = 2$.

2.3 Endpoint constants

Write $L = \log q$; all logarithms are natural. The elementary bounds used below are proved in Appendix A. In particular,

$$\frac{207}{20} < \log q_0 < \frac{21}{2}, \qquad \sqrt{q_0} > 180. \tag{4}$$

Indeed $q_0 = 2^{15}$, $69/100 < \log 2 < 7/10$, and $180^2 < 32768$. Since $(\log q)/\sqrt{q}$ decreases when $\log q > 2$, for every $q \geq q_0$ we have

$$L > 207/20, \qquad L/u < 7/120. \tag{5}$$

3 Elementary prime estimates

Write

$$\vartheta(x) = \sum_{p \leq x} \log p, \qquad A(x) = \sum_{p \leq x} \frac{\log p}{p}, \qquad B = \sum_{p} \frac{\log p}{p^2}.$$

A letter p under a sum always denotes a rational prime. The following estimates suffice; no lower bound for ϑ is required.

Lemma 3. *For $x \geq 1$ and integers $m \geq 2$,*

$$\vartheta(x) < \frac{3}{2}x, \qquad \log m - \frac{7}{3} < A(m) < \log m + \frac{3}{2}, \qquad B < \frac{2}{3}. \tag{6}$$

Proof. Write $n = 2k$ or $n = 2k+1$. Every prime in $(\lceil n/2 \rceil, n]$ divides the central binomial coefficient $\binom{n}{\lfloor n/2 \rfloor}$. This coefficient is at most 2^{2k}: in the odd case there are two equal central coefficients. Thus

$$\vartheta(n) - \vartheta(\lceil n/2 \rceil) \leq 2\lfloor n/2 \rfloor \log 2.$$

Starting at $n = 1$, induction gives $\vartheta(n) < 2n \log 2$. Taking $n = \lfloor x \rfloor$, and using $\log 2 < 7/10 < 3/4$, proves the first assertion.

Every prime other than 2 is odd, and $f(x) = \log x / x^2$ decreases for $x \geq 3$. For each odd integer $n \geq 5$, $f(n) \leq \frac{1}{2} \int_{n-2}^{n} f(x)\, dx$. Hence

$$\begin{aligned} B &\leq \frac{\log 2}{4} + \frac{\log 3}{9} + \frac{1}{2} \int_3^\infty \frac{\log x}{x^2}\, dx \\ &= \frac{\log 2}{4} + \frac{5}{18} \log 3 + \frac{1}{6} < \frac{233}{360} < \frac{2}{3}, \end{aligned}$$

using $\log 3 < 11/10$.

Counting the multiples of $p, p^2, \dots$ among $1, \dots, m$ gives $v_p(m!) = \sum_{j\ge1} \lfloor m/p^j \rfloor$, where v_p denotes the exponent of p in an integer. Thus integral comparison and $1/(p(p-1)) \le 2/p^2$ give

$$m \log m - m < \log(m!) \le mA(m) + m \sum_p \frac{\log p}{p(p-1)} \le mA(m) + 2mB < mA(m) + \frac{4}{3}m.$$

This proves the lower bound for $A(m)$. In the other direction, $\log(m!) \ge mA(m) - \vartheta(m)$ and $\log(m!) \le m \log m$ prove the upper bound. □

4 Norm counts and an auxiliary inert prime

Counting norm representations will bound the number of split primes and yield an auxiliary inert prime.[2] We use an unsieved norm count and a weighted interval of ordinary primes.

Let $\pi_+(x)$ count split rational primes at most x. For each integer $b > 0$, the real interval of a satisfying $Q(a, b) \le x$ has length $2\sqrt{x - qb^2/4}$, when nonempty. Bounding the number of integers by length plus 1, and the decreasing length sum by its integral, gives

$$\#\{(a, b) \in \mathbb{Z} \times \mathbb{Z}_{>0} : Q(a, b) \le x\} \le \frac{\pi x}{u} + \frac{2\sqrt{x}}{u}.$$

Every split prime p is represented by Q, as proved in Section 2. Choose a representing $z = a + b\omega$ with $b > 0$. Then $-\bar{z}$ is another such point. They are distinct: equality would give $2a + b = 0$, hence $b = 2k$ and $p = qk^2$, impossible for a split prime. Thus each split prime supplies at least two points with $b > 0$, and consequently

$$\pi_+(x) \le \frac{\pi x}{2u} + \frac{\sqrt{x}}{u}. \tag{7}$$

For $X \ge Y \ge q/4$, partial summation, followed by (7), yields

$$\sum_{\substack{Y < p \le X \\ p \text{ split}}} \frac{\log p}{p} \le \frac{\pi}{4u}\big((\log X)^2 - (\log Y)^2 + 2 \log Y\big) + \frac{2(\log Y + 1)}{u\sqrt{Y}}. \tag{8}$$

For clarity, this is only an identity obtained by summing $f(p) = f(X) - \int_p^X f'(t)\, dt$, with $f(t) = \log t/t$, over a finite set of primes and subtracting the contribution below Y. It gives

$$\frac{\log X}{X}\pi_+(X) - \frac{\log Y}{Y}\pi_+(Y) + \int_Y^X \pi_+(t)\frac{\log t - 1}{t^2}\, dt.$$

Discard the nonpositive lower boundary term. Substitution of (7) then gives the right side of (8), together with the further nonpositive term $-(\log X + 2)/(u\sqrt{X})$.

Put $S_+(X) = \sum_{p \le X,\ p \text{ split}} \log p/p$. There are no split primes at or below $q/4$, by Section 2 and the fact that $q/4$ is not an integer. Taking $Y = q/4$ in (8), and using $\log(q/4) > 2$ and $\pi < 16/5$, gives

$$S_+(X) \le \frac{4}{5u}(\log X)^2 + \frac{4(L + 1)}{q} \qquad (X \ge q/4). \tag{9}$$

Lemma 4. *There is an inert prime R with $q^4 < R \le q^5$.*

[2] *Modern attribution.* Pratt [M4] suggested this combination of norm counts and a supply of inert primes. The estimates needed here are derived below.

Proof. Lemma 3 shows that the total weight $\log p/p$ of primes in $(q^4, q^5]$ is greater than $L - 23/6$. By (8), the split-prime weight there is at most

$$\frac{4}{5u}(9L^2 + 8L) + \frac{2(4L+1)}{uq^2}.$$

It is therefore enough to check

$$\frac{23}{6L} + \frac{4}{5u}(9L + 8) + \frac{2(4 + 1/L)}{uq^2} < 1. \tag{10}$$

Each term decreases as q increases in our range. For the middle term, differentiation with respect to L gives a factor $9 - (9L + 8)/2 < 0$; the other terms plainly decrease. At $q = q_0$, (4) bounds the left side by

$$\frac{10}{27} + \frac{41}{90} + \frac{2(4 + 20/207)}{180 \cdot 32768^2} < \frac{10}{27} + \frac{41}{90} + \frac{1}{1000} < \frac{5}{6}.$$

The only ramified prime, q, is below the interval. Since the split primes cannot account for its full weight, an inert prime exists. □

5 The derivative determinant

We construct a determinant of polynomial derivatives evaluated on a rectangular set of elements of $\mathcal{A}$.[3] Choose R as in Lemma 4 and put

$$S = \lfloor R/12 \rfloor, \qquad J = \lceil R/48 \rceil, \qquad N = RS, \qquad T = 2\sqrt{Ru}.$$

Take all points

$$z = a + b\omega, \qquad |a| \le A_0 := \lceil \sqrt{Ru} \rceil, \qquad |b| \le B_0 := \lceil \sqrt{R/u} \rceil.$$

Their number is $n = (2A_0 + 1)(2B_0 + 1) \ge 4R$, so $nJ \ge RS$. Since $|\omega| \le 3u/5$, $u > 1$, and $R > q^4 > 100u$,

$$|z| \le \frac{8}{5}\sqrt{Ru} + 1 + \frac{3}{5}u \le \frac{44}{25}\sqrt{Ru} < T.$$

Here $\sqrt{Ru} > 10u$. Also $2T < R$ and $1 < T < R$.

Form the matrix whose rows are indexed by (r, s), $0 \le r < R$, $0 \le s < S$, and whose columns are indexed by a point z_i and an order $0 \le k < J$. The entry is

$$\partial_z^k (z^r w^s)\Big|_{(z_i, \bar{z}_i)}. \tag{11}$$

These are ordinary derivatives, without division by factorials.

5.1 Nonvanishing over a finite field

We prove nonvanishing by reducing the array modulo the auxiliary prime.[4]

Reduce (11) modulo the inert prime R, in the quadratic arithmetic of Section 2. This field has R^2 elements. The nodes remain distinct: every nonzero element of $R\mathcal{A}$ has modulus at least R, by

[3] *Modern attribution.* This construction is adapted from Green [M1, §2].

[4] *Modern attribution.* This argument follows van Dijk [M8], as highlighted by Green [M2]. Green also attributes essentially the same method to unpublished work of Jacob Pescador. Our source is the public argument.

the positive integral norm proved in Section 2, whereas any difference of two nodes has modulus less than $2T < R$. Equation (3) gives $\bar{z}_i = z_i^R$ in this field.

If a nonzero row combination vanished, let $F(z, w)$ be its polynomial and substitute $w = z^R$. The exponents $r + Rs$ are distinct, so $F(z, z^R)$ is nonzero and has degree at most $RS - 1$. Its derivatives agree with the prescribed partial derivatives, because the derivative of z^R is zero in characteristic R. Since $J \le R$ and $k! \not\equiv 0$ for $0 \le k < J$, each node is a root of multiplicity at least J: the coefficient of $(z - z_i)^k$ in the translated polynomial is its k-th derivative at z_i, divided by $k!$. Successive polynomial division then forces its degree to be at least $nJ \ge RS$, a contradiction.

The rows are therefore linearly independent over this field. Ordinary elimination (using only field operations) selects N columns with nonzero determinant modulo R. Fix this nonzero N-square minor $D_0 \in \mathcal{A}$, and set

$$\Delta = D_0 \overline{D_0} \in \mathbb{Z}_{>0}.$$

Only this minor is used from now on.

5.2 The complex absolute-value bound

Each entry in row (r, s) has modulus at most $T^{r+s}(R/T)^J$: the derivative contributes at most R^k, and an entry with $k > r$ is zero. Expand the determinant as the signed sum of its $N!$ permutation products, and use the triangle inequality and $N! \le N^N$. This elementary determinant algebra is documented in Cauchy [H2, Chapter III, §2]. We obtain

$$|D_0| \le N! \prod_{r,s} T^{r+s}(R/T)^J,$$

and hence, after squaring and summing $r + s$,

$$\frac{\log \Delta}{R^2 S} \le \frac{2 \log N}{R} + \left(1 + \frac{S}{R}\right) \log T + \frac{2J}{R} \log(R/T).$$

Now $S/R \le 1/12$, $J/R \le 1/48 + 1/R$, and $N < R^2$. Thus the coefficient of $\log T$, after combining the main terms, is $25/24$. Since $\log T = \log 2 + \frac{1}{2} \log R + \frac{1}{4} L$, we obtain

$$\frac{\log \Delta}{R^2 S} \le \frac{9}{16} \log R + \frac{25}{96} L + \frac{25}{24} \log 2 + \frac{6 \log R}{R}. \tag{12}$$

5.3 Divisibility at the inert primes

A triangular change of row polynomials will force divisibility at each inert prime.[5] We give the full argument, including $p = 2$. For an inert prime p, put

$$h_{r,s} = \min(\lfloor r/p \rfloor, S - 1 - s)$$

and replace each row polynomial $z^r w^s$ by

$$z^{r - p h_{r,s}} (z^p - w)^{h_{r,s}} w^s.$$

The leading term is $z^r w^s$, and every other term has smaller z-degree while remaining in the rectangle of row indices. Ordering rows by z-degree makes this an integer triangular transformation of determinant one.

[5] *Modern attribution.* The transformation is adapted from Green [M1, Lemma 5.1]; Pratt [M4] suggested using this stronger divisibility estimate.

At each node $z_i^p - \bar{z}_i \in p\mathcal{A}$, by (3). Every positive ordinary derivative of $z^p - w$ is also divisible by p. By the product rule, all entries in the transformed row are therefore divisible by $p^{h_{r,s}}$. Taking norms gives

$$v_p(\Delta) \geq 2 \sum_{r,s} h_{r,s}. \tag{13}$$

The row transformation may depend on p; each divisibility is nevertheless a statement about the same integer Δ.

We bound this sum with generous constants. Direct summation gives, for every integer $k \geq 0$,

$$\sum_{s<S} \min(k, S-1-s) \geq Sk - \frac{k(k+1)}{2}.$$

Writing $R = ap + b$, $0 \leq b < p$, gives

$$\sum_{r<R} \lfloor r/p \rfloor = \frac{R(R-p) + b(p-b)}{2p} \geq \frac{R(R-p)}{2p}.$$

Also $\lfloor r/p \rfloor (\lfloor r/p \rfloor + 1) \leq r^2/p^2 + r/p$. Substituting these estimates yields

$$\frac{2\sum_{r,s} h_{r,s}}{R^2 S} \geq \left(1 - \frac{R-1}{2RS}\right)\frac{1}{p} - \frac{1}{R} - \frac{(R-1)(2R-1)}{6RSp^2}. \tag{14}$$

Let $I = \sum_{p \leq R,\ p \text{ inert}} \log p/p$. Summing (14), weighted by $\log p$, and using $S \geq R/13$ and Lemma 3, we find

$$\begin{aligned} \frac{\log \Delta}{R^2 S} &\geq I - \frac{I}{2S} - \frac{\vartheta(R)}{R} - \frac{R}{3S} B \\ &> I - \frac{I}{2S} - \frac{3}{2} - \frac{26}{9} = I - \frac{I}{2S} - \frac{79}{18}. \end{aligned} \tag{15}$$

Here $S \geq R/12 - 1 \geq R/13$, since $R > 156$.

6 One endpoint comparison

Separating the three splitting types in $A(R)$, and noting that the only ramified contribution is L/q, gives

$$I > \log R - \frac{7}{3} - S_+(R) - \frac{L}{q}, \qquad I < \log R + \frac{3}{2}.$$

Combine this with (12), (15), and (9). If

$$G(x) = \frac{7}{16}x - \frac{25}{96}L - \frac{4}{5u}x^2,$$

the resulting necessary inequality is

$$G(\log R) < \frac{121}{18} + \frac{25}{24}\log 2 + E, \qquad E = \frac{5L+4}{q} + \frac{I}{2S} + \frac{6\log R}{R}. \tag{16}$$

All boundary and rounding terms are included in E. In fact

$$E < \frac{5L+4}{q} + \frac{(25/2)\log R + 39/4}{R} < \frac{113}{65536} + \frac{2039}{4 \cdot 10^{16}} < \frac{1}{500}. \tag{17}$$

For the second inequality, the first function decreases with q in our range, so use (4) at $q = q_0$. The second decreases with R; use $R > q_0^4 > 10^{16}$ and $\log(10^{16}) < 40$. Thus $\log 2 < 7/10$ gives

$$G(\log R) < \frac{121}{18} + \frac{35}{48} + \frac{1}{500} < \frac{15}{2}. \tag{18}$$

On the other hand, $4L < \log R \le 5L$ by Lemma 4, and G is concave. Its minimum on $[4L, 5L]$ is therefore at an endpoint. By (5),

$$G(5L) - G(4L) = L\left(\frac{7}{16} - \frac{36}{5}\frac{L}{u}\right) > \frac{7}{400}L > 0.$$

It follows that

$$G(\log R) \ge G(4L) = \frac{143}{96}L - \frac{64}{5}\frac{L^2}{u}.$$

Together with (18), this forces

$$\frac{143}{96} < \frac{64}{5}\frac{L}{u} + \frac{15}{2L} < \frac{56}{75} + \frac{50}{69} = \frac{846}{575} < \frac{143}{96},$$

a contradiction. This proves Theorem 1.

7 The remaining finite range: a printed certificate

We now drop the assumption $q \ge 32768$. Theorem 1 and (2) reduce the exclusion problem to $8 < q < 32768$, $q \equiv 3 \pmod 8$. We exclude the remaining candidates by explicit small-prime congruences.[6] For any odd prime p and integer a,

$$4p < q \quad \text{and} \quad p \mid (q + a^2) \quad \Longrightarrow \quad h(-q) \ne 1 \tag{19}$$

when $-q$ is fundamental. The tables may include nonfundamental values; excluding those extra values causes no problem.

7.1 The thirty residue classes above 163

For $q > 163$, every prime at most 31 satisfies $4p < q$. The condition that $-q$ be a nonzero nonsquare, first at $3, 5, 7, 11$, leaves exactly

$$\begin{gathered} q \equiv 3 \pmod 8, \qquad q \bmod 3 \in \{1\}, \qquad q \bmod 5 \in \{2, 3\}, \\ q \bmod 7 \in \{1, 2, 4\}, \qquad q \bmod 11 \in \{1, 3, 4, 5, 9\}. \end{gathered} \tag{20}$$

The Chinese remainder theorem gives exactly $1 \cdot 1 \cdot 2 \cdot 3 \cdot 5 = 30$ residue classes modulo $M = 8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 9240$. For completeness, this use of the theorem requires only Bezout's identity: for coprime m, n, the numbers $a + mk$, $0 \le k < n$, run through all residues modulo n; iterate. The thirty distinct representatives $0 \le r < M$ in Table 2 each satisfy (20), so they are the complete list.

Every remaining candidate is $q = r + 9240t$ with $t \in \{0, 1, 2, 3\}$, since $4M > 32768$. In the column for t, an entry $p : a$ certifies that $q + a^2$ is divisible by p. All the listed p's are primes between 13 and 31. A dash means $q \le 163$ or $q \ge 32768$. Thus the table has a witness for every candidate in $163 < q < 32768$. It contains only 104 witnesses. For example, the entry $29 : 8$ in the row 2347, column 3, checks

$$2347 + 3 \cdot 9240 + 8^2 = 30131 = 29 \cdot 1039.$$

This completes the exclusion above 163 by integer arithmetic.

[6] *Modern attribution.* This finite-sieve strategy was proposed by Pratt [M3].

Table 2: Complete certificate for $163 < q < 32768$, with $q = r + 9240t$. Entries are $p : a$; see (19).

r	$t = 0$	$t = 1$	$t = 2$	$t = 3$
67	-	13:1	13:2	17:5
163	-	13:3	13:5	19:3
883	13:1	13:2	17:0	13:6
907	13:4	17:6	13:3	13:5
1387	13:2	17:7	13:6	13:0
1747	17:2	29:2	13:1	13:2
2083	13:6	13:0	13:4	17:7
2227	13:3	13:5	17:4	19:4
2347	17:4	13:3	13:5	29:8
3067	13:1	13:2	17:3	13:6
3403	13:4	19:7	13:3	13:5
3523	13:0	13:4	23:10	13:3
3763	23:3	13:6	13:0	13:4
4027	13:4	23:2	13:3	13:5
4603	13:5	23:7	23:3	31:14
4867	19:4	29:4	13:1	13:2
5083	13:0	13:4	17:4	-
5107	19:2	17:1	17:3	-
5443	13:2	19:2	13:6	-
5707	13:0	13:4	17:2	-
5923	19:9	17:1	17:3	-
6043	17:3	17:0	17:5	-
6427	17:4	29:14	13:1	-
6763	13:6	13:0	13:4	-
7627	13:2	19:9	13:6	-
7723	13:5	19:2	19:6	-
8563	13:2	17:8	13:6	-
8803	23:11	13:1	13:2	-
8947	13:6	13:0	13:4	-
9067	23:8	13:6	13:0	-

7.2 The twenty candidates up to 163

Among $q = 11, 19, \ldots, 163$, the fifteen values in Table 3 have witnesses satisfying (19). The only survivors are $11, 19, 43, 67, 163$. The negative fundamental discriminants with $q \leq 8$ have absolute values $3, 4, 7, 8$. This proves the forward implication of Theorem 2.

q	$p : a$
27	3:0
35	3:1
51	3:0
59	3:1
75	3:0
83	3:1
91	5:2
99	3:0
107	3:1
115	5:0
123	3:0
131	3:1
139	5:1
147	3:0
155	3:1

Table 3: The fifteen excluded values with $11 \leq q \leq 163$.

7.3 Checking that the nine survivors have class number one

Here is a short reduction argument, in the setting of Gauss [H1, Section V]. In any proper equivalence class, choose a form with the least possible positive leading coefficient A. An integral substitution $x \mapsto x + ky$ then arranges $-A < B \leq A$. The determinant-one substitution $(x, y) \mapsto (-y, x)$ shows that $A \leq C$, by minimality. If $A = C$, the same substitution permits $B \geq 0$. Consequently every class has a representative satisfying

$$|B| \leq A \leq C, \qquad B \geq 0 \text{ if } |B| = A \text{ or } A = C.$$

Such a representative has $q = 4AC - B^2 \geq 3A^2$. For each of our nine values, check $1 \leq A \leq \lfloor \sqrt{q/3} \rfloor$, $-A \leq B \leq A$, retaining only

$$C = (B^2 + q)/(4A) \in \mathbb{Z}, \quad C \geq A, \quad \gcd(A, B, C) = 1,$$

and the boundary convention just stated. Table 4 records all the resulting triples; the largest A to check is 7. Each discriminant has just one such form, so every class is its class. This proves the converse, without a theorem about ideal classes or uniqueness of reduction.

q	$\lfloor\sqrt{q/3}\rfloor$	All reduced triples (A, B, C)
3	1	$(1, 1, 1)$
4	1	$(1, 0, 1)$
7	1	$(1, 1, 2)$
8	1	$(1, 0, 2)$
11	1	$(1, 1, 3)$
19	2	$(1, 1, 5)$
43	3	$(1, 1, 11)$
67	4	$(1, 1, 17)$
163	7	$(1, 1, 41)$

Table 4: A finite check with $A \leq 7$ establishes the converse.

Reproducibility and status

The paper, including its tables and Appendix A, is the proof. Supplementary Python code checks the printed tables directly in the LaTeX source, the rational inequalities, and the nine reduced-form lists. It also cross-checks the certificate by a direct sieve of all 4095 candidates $q \equiv 3 \pmod 8$, $8 < q < 32768$. No determinant of the enormous size occurring in the existence argument needs to be evaluated. The code uses only the Python standard library:

```
python3 verify_constants.py
python3 verify_certificate.py
```

These are exact arithmetic checks supporting a mathematical proof, not formal verification. They do not test the historical eligibility of its concepts. The mathematical argument and the proposed 1830 cutoff require separate review; no independent audit verdict is asserted for this draft.

Acknowledgements and AI disclosure

The author thanks Tao and Green for the historical question, and the participants in Tao's discussion for their public arguments and suggestions. The mathematical contributions used here are cited where they enter the proof. This note and its supplementary code were prepared with OpenAI Codex. Green [M1] credits ChatGPT for the mathematical arguments in his source paper; van Dijk [M8] credits GPT-6 Astra and Opus 5.5 for his new arguments.

A Elementary bounds for the numerical constants

For $0 < t < 1$, integration of a finite geometric series, with its remainder bounded by a geometric tail, gives

$$\log\frac{1+t}{1-t} = 2\sum_{j=0}^{m-1}\frac{t^{2j+1}}{2j+1} + E_m, \qquad 0 < E_m < \frac{2t^{2m+1}}{(2m+1)(1-t^2)}.$$

At $t = 1/3$, use two terms for a lower bound and one term plus the tail for an upper bound. At $t = 1/2$, use two terms plus the tail. These give

$$\frac{69}{100} < \frac{56}{81} < \log 2 < \frac{25}{36} < \frac{7}{10}, \qquad \log 3 < 1 + \frac{1}{12} + \frac{1}{60} = \frac{11}{10}.$$

Consequently $\log 10 < \log 12 = \log 3 + 2\log 2 < 5/2$.

For the harmless geometric constant, the tangent addition identity gives $\pi/4 = \arctan(1/2) + \arctan(1/3)$. Integrating $1/(1+x^2) = 1 - x^2 + x^4 - x^6/(1+x^2)$ gives $\arctan x < x - x^3/3 + x^5/5$, for $x > 0$. Therefore

$$\pi < 4\left(\frac{1}{2} - \frac{(1/2)^3}{3} + \frac{(1/2)^5}{5} + \frac{1}{3} - \frac{(1/3)^3}{3} + \frac{(1/3)^5}{5}\right) = \frac{6115}{1944} < \frac{16}{5}.$$

All remaining comparisons in the proof are between displayed rational numbers. For example, the final margin is

$$\frac{143}{96} - \left(\frac{56}{75} + \frac{50}{69}\right) = \frac{1009}{55200} > 0.$$

Historical sources

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Modern sources and attribution

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